Highlights

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ABSTRACT

Here goes the abstract Lo pongo aqui no se muy bien por que. Creo que el enfoque adecuado para el paper debe ser el de, especificar que en casos donde los datos se presenten únicamente en forma de nº de carga y nº de descarga, mientras que la solucion general suele ser el representar al pasajero como agente e inferir su deseo mediante una OD matrix, al tener estas restricciones de datos junto a que el objetivo no es analizar el comportamiento de los pasajeros, sino teniendo en cuenta el número de pasajeros, analizar cómo se comportan los autobuses, un enfoque que no use OD, que se suele estimar por IPF (con busqueda de maxima entropia y que DEPENDE MUCHO de la matriz semilla inicial, normalmente obtenida de encuestas o datos de tarjeta), se propone un enfoque donde la carga y descarga se infiere de forma independiente. Esto se suma a, debido a que no se busca estudiar el comportamiento particular de pasajeros concretos, se puede reducir la granularidad al cargarnos el agente concreto, al no requerir de procesos coginitivos en él que alteren el comportamiento.

1. Introduction

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2. State of the Art

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Table 1
Specifications of route Express 01

Parameter	Value
Directions	2
Direction label 1	Cibeles
Direction label 2	La Peseta
Cibeles number of stops	15
La Peseta number of stops	12

Table 2
Fixed simulation parameters

Parameter	Value
Line	E01 (Cibeles-La Peseta)
Total physical buses	12
Total trips	163
Initial TimeSlot	5
Extra dwell time per passenger	5
Simulation time step	2

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3. Proposed solutions: two different approaches

3.1. Agents and protocols

4. Implementations and results

Both proposed architectures run in the dedicated agent platform GAMA **Cita** . The election is based on both the use of dedicated FIPA inboxing, allowing implementation-defined conversations, and a headless batch mode that improves performance on the simulations.

4.1. Simulation parameters

For the simulation characterization, the layout of the route Express 01 has been used. The route specs are presented in table 1.

The specific parameterization used for the simulation is shown in table 2. The field `Extra dwell time per passenger` is used with both a premise and a comparasion.

First, it assumes that the boarding and alightment processes runs in parallel, as in regular bus operations the processes runs in different doors.

Secondly, the dwell extra time (E) will be the result of the following equation:

$$E = \max\big(\max(b,\, a)\, t_p,\; E_{\min}\big) \tag{1}$$

where b and a are the numbers of boarding and alighting passengers at the stop, t_p is the extra dwell time per passenger, and $E_{\min}$ is the minimum dwell imposed by the interaction protocol.

As measurement for converging randominess, the Monte Carlo method is implemented, specifying 1000 replicas per simulation.

4.2. Results

5. Conclusions

6.

CRediT authorship contribution statement

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References

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